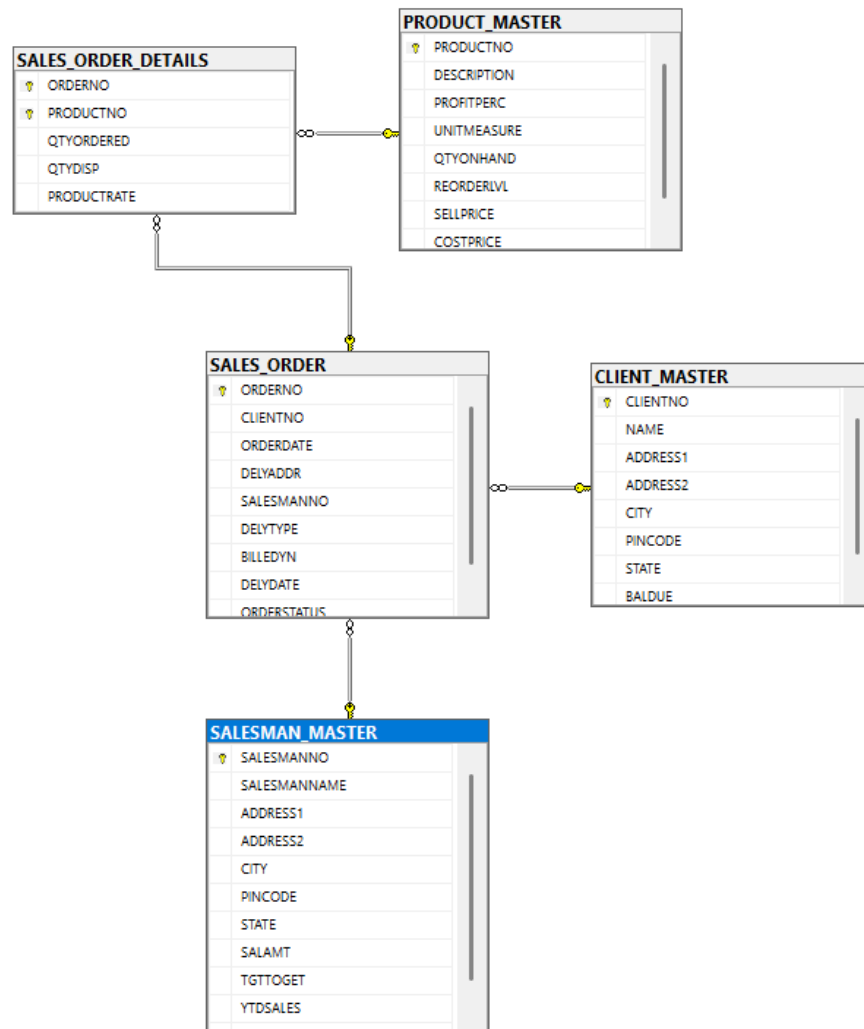


Weekly Assignment-1

SQL and PL/SQL

1) Write CREATE TABLE commands for the following table descriptions:



--Database Creation

```
CREATE DATABASE Sales;
```

--Using Sales Database

```
USE Sales;
```

--Table Creation Commands

--CLIENT_MASTER Table Creation

```
CREATE TABLE CLIENT_MASTER(  
    CLIENTNO VARCHAR(6) PRIMARY KEY,  
    NAME VARCHAR(20) NOT NULL,  
    ADDRESS1 VARCHAR(30),  
    ADDRESS2 VARCHAR(30),  
    CITY VARCHAR(15),  
    PINCODE NUMERIC(8),  
    STATE VARCHAR(15),  
    BALDUE NUMERIC(10,2),  
    CONSTRAINT chk_clientno CHECK (CLIENTNO LIKE 'C%')  
)
```

--PRODUCT_MASTER Table Creation

```
CREATE TABLE PRODUCT_MASTER(  
    PRODUCTNO VARCHAR(6) PRIMARY KEY,  
    DESCRIPTION VARCHAR(15) NOT NULL,  
    PROFITPERC NUMERIC(4,2) NOT NULL,  
    UNITMEASURE VARCHAR(10) NOT NULL,  
    QTYONHAND NUMERIC(8) NOT NULL,  
    REORDERLVL NUMERIC(8) NOT NULL,  
    SELLPRICE NUMERIC(8,2) NOT NULL,  
    COSTPRICE NUMERIC(8,2) NOT NULL,  
    CONSTRAINT chk_productno CHECK (PRODUCTNO LIKE 'P%'),  
    CONSTRAINT chk_sellprice CHECK (SELLPRICE>0),  
    CONSTRAINT chk_costprice CHECK (COSTPRICE>0)  
);
```

--SALESMAN_MASTER Table Creation

```
CREATE TABLE SALESMAN_MASTER(  
    SALESMANNO VARCHAR(6) PRIMARY KEY,  
    SALESMANNAME VARCHAR(20) NOT NULL,  
    ADDRESS1 VARCHAR(30) NOT NULL,  
    ADDRESS2 VARCHAR(30),  
    CITY VARCHAR(20),  
    PINCODE NUMERIC(8),  
    STATE VARCHAR(20),  
    SALAMT NUMERIC(8,2) NOT NULL,  
    TGTTOGET NUMERIC(6,2) NOT NULL,  
    YTDSALES NUMERIC(6,2) NOT NULL,  
    REMARKS VARCHAR(60),  
    CONSTRAINT chk_salesmanno CHECK (SALESMANNO LIKE 'S%'),  
    CONSTRAINT chk_salamat CHECK (SALAMT>0)  
);
```

--SALES_ORDER Table Creation

```
CREATE TABLE SALES_ORDER(  
    ORDERNO VARCHAR(6) PRIMARY KEY,  
    CLIENTNO VARCHAR(6) NOT NULL,  
    ORDERDATE DATE,  
    DELYADDR VARCHAR(25),  
    SALESMANNO VARCHAR(6),  
    DELYTYPE CHAR(1),  
    BILLEDYN CHAR(1),  
    DELYDATE DATE,  
    ORDERSTATUS VARCHAR(10),  
  
    CONSTRAINT chk_orderno CHECK (ORDERNO LIKE 'O%'),  
  
    CONSTRAINT fk_client FOREIGN KEY (CLIENTNO)  
    REFERENCES CLIENT_MASTER(CLIENTNO),  
  
    CONSTRAINT fk_salesman FOREIGN KEY (SALESMANNO)  
    REFERENCES SALESMAN_MASTER(SALESMANNO),  
  
    CONSTRAINT chk_delytype CHECK (DELYTYPE IN ('P','F')),  
    CONSTRAINT chk_billed CHECK (BILLEDYN IN ('Y','N')),  
    CONSTRAINT chk_status CHECK (ORDERSTATUS IN ('In  
Process', 'Fulfilled', 'Backorder', 'Cancelled'))  
);
```

--SALES_ORDER_DETAILS Table Creation

```
CREATE TABLE SALES_ORDER_DETAILS(  
    ORDERNO VARCHAR(6),  
    PRODUCTNO VARCHAR(6),  
    QTYORDERED NUMERIC(8),  
    QTYDISP NUMERIC(8),  
    PRODUCTRATE NUMERIC(10,2),  
  
    CONSTRAINT fk_order FOREIGN KEY (ORDERNO)  
    REFERENCES SALES_ORDER(ORDERNO),  
  
    CONSTRAINT fk_product FOREIGN KEY (PRODUCTNO)  
    REFERENCES PRODUCT_MASTER(PRODUCTNO),  
  
    CONSTRAINT pk_order_product PRIMARY KEY (ORDERNO,PRODUCTNO)  
);
```

2) Insertion Commands

--Insertion Of Data Records into the Tables

--Inserting Data into Client_Master

```
INSERT INTO Client_Master  
(CLIENTNO, NAME, ADDRESS1, ADDRESS2, CITY, PINCODE, STATE, BALDUE)  
VALUES  
( 'C00001', 'Ivan Bayross', 'A/1', 'Street 1', 'Mumbai', 400054, 'Maharashtra', 15000),  
( 'C00002', 'Ravi Kumar', 'B/2', 'Street 5', 'Pune', 411001, 'Maharashtra', 12000),  
( 'C00003', 'Riya Shah', 'C/7', 'Ring Road', 'Ahmedabad', 380001, 'Gujarat', 8000),  
( 'C00004', 'Neeraj Verma', 'H/9', 'CP Area', 'Delhi', 110001, 'Delhi', 22000),  
( 'C00005', 'Sneha Patil', '10/2', 'MG Road', 'Nashik', 422001, 'Maharashtra', 9500),  
( 'C00006', 'Amit Mehta', 'D/44', 'City Center', 'Surat', 395001, 'Gujarat', 17500),  
( 'C00007', 'Priya Nair', 'P/22', 'Marine Drive', 'Kochi', 682001, 'Kerala', 11000),  
( 'C00008', 'Rahul Jain', 'K/17', 'Pink Road', 'Jaipur', 302001, 'Rajasthan', 14000),  
( 'C00009', 'Aditya Singh', 'G/8', 'Hazratganj', 'Lucknow', 226001, 'Uttar  
Pradesh', 20000),  
( 'C00010', 'Divya Kapoor', 'M/3', 'Beach Road', 'Chennai', 600001, 'Tamil Nadu', 9000),  
( 'C00011', 'Arvind Kapoor', 'M/3', 'Beach Road', 'Mumbai', 400001, 'Maharashtra', 9000);
```

--Inserting Data into Product_Master

```
INSERT INTO Product_Master  
(PRODUCTNO, DESCRIPTION, PROFITPERC, UNITMEASURE, QTYONHAND, REORDERLVL, SELLPRICE, COST  
PRICE)  
VALUES  
( 'P00001', 'T-Shirts', 5, 'Piece', 200, 50, 350, 250),  
( 'P00002', 'Jeans', 10, 'Piece', 150, 30, 1200, 900),  
( 'P00003', 'Jackets', 15, 'Piece', 80, 20, 2500, 1800),  
( 'P00004', 'Shoes', 12, 'Pair', 120, 25, 3000, 2100),  
( 'P00005', 'Caps', 8, 'Piece', 300, 60, 250, 150),  
( 'P00006', 'Belts', 6, 'Piece', 220, 40, 450, 300),  
( 'P00007', 'Socks', 7, 'Pair', 400, 100, 150, 80),  
( 'P00008', 'Shirts', 9, 'Piece', 180, 40, 900, 650),  
( 'P00009', 'Shorts', 11, 'Piece', 160, 35, 700, 450),  
( 'P00010', 'Sweaters', 14, 'Piece', 70, 15, 2800, 2000),  
( 'P00011', '1.44 drive', 10, 'Piece', 50, 10, 1100.50, 900);  
( 'P00012', 'Trousers', 12, 'Piece', 150, 40, 1500, 1100),  
( 'P00013', 'Pull Overs', 12, 'Piece', 120, 30, 1800, 1300);
```

--Inserting Data into Salesman_Master

INSERT INTO Salesman_Master

(SALESMANNO, SALESMANNAME, ADDRESS1, ADDRESS2, CITY, PINCODE, STATE,
SALAMT, TGTTOTET, YTDSALES, REMARKS)

VALUES

('S00001', 'Aman', 'A/14', 'Worli', 'Mumbai', 400002, 'Maharashtra', 3000, 100, 50, 'Good'),
(
'S00002', 'Rohit', '101', 'Andheri', 'Mumbai', 400053, 'Maharashtra', 4500, 150, 90, 'Excellent'),
(
'S00003', 'Neha', '22B', 'Bandra', 'Mumbai', 400050, 'Maharashtra', 3800, 120, 85, 'Good Performer'),
(
'S00004', 'Kunal', 'Flat
9', 'Parel', 'Mumbai', 400012, 'Maharashtra', 4200, 140, 95, 'Very Good'),
(
'S00005', 'Meera', 'H/7', 'Dadar', 'Mumbai', 400014, 'Maharashtra', 3900, 110, 88, 'Consistent'),
(
'S00006', 'Raj', 'B/21', 'Thane', 'Mumbai', 400601, 'Maharashtra', 4600, 160, 100, 'Star Performer'),
(
'S00007', 'Ankit', 'C/3', 'Kandivali', 'Mumbai', 400067, 'Maharashtra', 3500, 105, 70, 'Average'),
(
'S00008', 'Pooja', 'D/12', 'Borivali', 'Mumbai', 400092, 'Maharashtra', 4100, 130, 92, 'Good'),
(
'S00009', 'Varun', '33A', 'Vashi', 'Navi
Mumbai', 400703, 'Maharashtra', 4800, 180, 110, 'Excellent'),
(
'S00010', 'Smita', '55C', 'Chembur', 'Mumbai', 400071, 'Maharashtra', 3700, 115, 78, 'Reliable');

--Inserting Data into Sales_Order

INSERT INTO Sales_Order

(ORDERNO, ORDERDATE, CLIENTNO, DELYTYPE, BILLEDYN, SALESMANNO, DELYDATE, ORDERSTATUS)

VALUES

('019001', '2002-06-12', 'C00001', 'F', 'N', 'S00001', '2002-07-20', 'In Process'),
(
'019002', '2002-07-15', 'C00002', 'P', 'Y', 'S00002', '2002-07-25', 'Fulfilled'),
(
'019003', '2002-08-01', 'C00003', 'F', 'N', 'S00003', '2002-08-10', 'Backorder'),
(
'019004', '2002-08-05', 'C00004', 'F', 'Y', 'S00004', '2002-08-12', 'Fulfilled'),
(
'019005', '2002-08-10', 'C00005', 'P', 'N', 'S00005', '2002-08-18', 'In Process'),
(
'019006', '2002-08-15', 'C00006', 'F', 'Y', 'S00006', '2002-08-22', 'Fulfilled'),
(
'019007', '2002-08-20', 'C00007', 'P', 'N', 'S00007', '2002-08-28', 'Cancelled'),
(
'019008', '2002-08-25', 'C00008', 'F', 'Y', 'S00008', '2002-09-02', 'Fulfilled'),
(
'019009', '2002-08-28', 'C00009', 'P', 'N', 'S00009', '2002-09-05', 'In Process'),
(
'019010', '2002-08-30', 'C00010', 'F', 'Y', 'S00010', '2002-09-10', 'Fulfilled');

--Inserting Data into Sales_Order_Details

INSERT INTO Sales_Order_Details

(ORDERNO, PRODUCTNO, QTYORDERED, QTYDISP, PRODUCTRATE)

VALUES

('019001', 'P00001', 4, 4, 525),
(
'019002', 'P00002', 2, 2, 1400),
(
'019003', 'P00003', 3, 1, 2700),
(
'019004', 'P00004', 1, 1, 3200),
(
'019005', 'P00005', 6, 4, 300),
(
'019006', 'P00006', 5, 5, 500),
(
'019007', 'P00007', 8, 6, 180),
(
'019008', 'P00008', 3, 3, 950),
(
'019009', 'P00009', 4, 2, 780),
(
'019010', 'P00010', 2, 2, 3000),
(
'019001', 'P00012', 2, 2, 1500),
(
'019002', 'P00012', 1, 1, 1500),
(
'019003', 'P00012', 3, 3, 1500),
(
'019004', 'P00002', 4, 4, 1200),
(
'019005', 'P00002', 1, 1, 1200),
(
'019002', 'P00001', 3, 3, 350),
(
'019002', 'P00013', 1, 1, 1800),
(
'019003', 'P00013', 3, 3, 1800);

Queries

(I) Answer following queries with the help of above schema:

--Answer following queries with the help of above schema :

1. Display the names of all the clients.

--1. Display the names of all the clients.

```
SELECT NAME FROM CLIENT_MASTER;
```

Output:

	NAME
1	Ivan Bayross
2	Ravi Kumar
3	Riya Shah
4	Neeraj Verma
5	Sneha Patil
6	Amit Mehta
7	Priya Nair
8	Rahul Jain
9	Aditya Singh
10	Diya Kapoor
11	Arvind Kapoor

2. Display all the clients who are located in Mumbai.

--2. Display all the clients who are located in Mumbai.

```
SELECT * FROM CLIENT_MASTER  
WHERE CITY='Mumbai';
```

Output:

	CLIENTNO	NAME	ADDRESS1	ADDRESS2	CITY	PINCODE	STATE	BALDUE
1	C00001	Ivan Bayross	A/1	Street 1	Mumbai	400054	Maharashtra	15000.00
2	C00011	Arvind Kapoor	M/3	Beach Road	Mumbai	400001	Maharashtra	9000.00

3. Display all the products whose selling price is > 2000 and < 5000.

--3. Display all the products whose selling price is >2000 and <5000.

```
SELECT * FROM PRODUCT_MASTER  
WHERE SELLPRICE>2000 AND SELLPRICE<5000;
```

Output:

	PRODUCTNO	DESCRIPTION	PROFITPERC	UNITMEASURE	QTYONHAND	REORDERLVL	SELLPRICE	COSTPRICE
1	P00003	Jackets	15.00	Piece	80	20	2500.00	1800.00
2	P00004	Shoes	12.00	Pair	120	25	3000.00	2100.00
3	P00010	Sweaters	14.00	Piece	70	15	2800.00	2000.00

4. Display Name, City and State of Clients not in the state of Maharashtra.

```
--4. Display Name, City and State of Clients not in the state of Maharashtra.  
SELECT NAME,CITY,STATE FROM CLIENT_MASTER  
WHERE STATE NOT LIKE 'Maharashtra';
```

Output:

	NAME	CITY	STATE
1	Riya Shah	Ahmedabad	Gujarat
2	Neeraj Verma	Delhi	Delhi
3	Amit Mehta	Surat	Gujarat
4	Priya Nair	Kochi	Kerala
5	Rahul Jain	Jaipur	Rajasthan
6	Aditya Singh	Lucknow	Uttar Pradesh
7	Divya Kapoor	Chennai	Tamil Nadu

5. Display all the information of client_no C0001 and C0002.

```
--5. Display all the information of client_no C00001 and C00002.  
SELECT * FROM CLIENT_MASTER  
WHERE CLIENTNO IN ('C00001','C00002');
```

Output:

	CLIENTNO	NAME	ADDRESS1	ADDRESS2	CITY	PINCODE	STATE	BALDUE
1	C00001	Ivan Bayross	A/1	Street 1	Mumbai	400054	Maharashtra	15000.00
2	C00002	Ravi Kumar	B/2	Street 5	Pune	411001	Maharashtra	12000.00

6. Change the selling price of '1.44 drive' to Rs. 1150.50.

```
--6. Change the selling price of '1.44 drive' to Rs. 1150.50.  
--Before Updation  
SELECT SELLPRICE AS Price_Before_Updation FROM PRODUCT_MASTER;  
  
--Updation  
UPDATE PRODUCT_MASTER  
SET SELLPRICE=1150.50  
WHERE DESCRIPTION='1.44 drive';  
  
--After Updation  
SELECT SELLPRICE AS Price_After_Updation FROM PRODUCT_MASTER;
```

Output:

	Price_Before_Updation
1	1100.50

	Price_After_Updation
1	1100.50

7. Delete the record of client_no C0005.

--7. Delete the record of client_no C0005.

--Deletion

```
ALTER TABLE SALES_ORDER  
DROP CONSTRAINT fk_client;
```

```
ALTER TABLE SALES_ORDER  
ADD CONSTRAINT fk_client  
FOREIGN KEY (CLIENTNO)  
REFERENCES CLIENT_MASTER(CLIENTNO)  
ON DELETE CASCADE;
```

```
DELETE FROM CLIENT_MASTER  
WHERE CLIENTNO='C00005';
```

--After Deletion

```
SELECT * FROM CLIENT_MASTER;
```

Output:

	CLIENTNO
1	C00001
2	C00002
3	C00003
4	C00004
5	C00006
6	C00007
7	C00008
8	C00009

8. Display the clients who stay in a city whose second letter is 'a'.

--8. Display the clients who stay in a city whose second letter is 'a'.

```
SELECT * FROM CLIENT_MASTER  
WHERE CITY LIKE '_a%';
```

Output:

	CLIENTNO	NAME	ADDRESS1	ADDRESS2	CITY	PINCODE	STATE	BALDUE
1	C00008	Rahul Jain	K/17	Pink Road	Jaipur	302001	Rajasthan	14000.00

9. Count the number of products having price greater than or equal to 1500.

```
--9. Count the number of products having price greater than or equal to 1500.  
SELECT COUNT(*) AS ProductCount_With_PriceMoreThan1499  
FROM PRODUCT_MASTER  
WHERE SELLPRICE>=1500;
```

Output:

Results Messages	
	ProductCount_With_PriceMoreThan1499
1	3

10. Display qtyordered, qtydisp and balancedqty (not in table).

```
--10. Display qtyordered, qtydisp and balancedqty (not in table).  
SELECT QTYORDERED,QTYDISP,(QTYORDERED-QTYDISP) AS BalancedQty  
FROM SALES_ORDER_DETAILS;
```

Output:

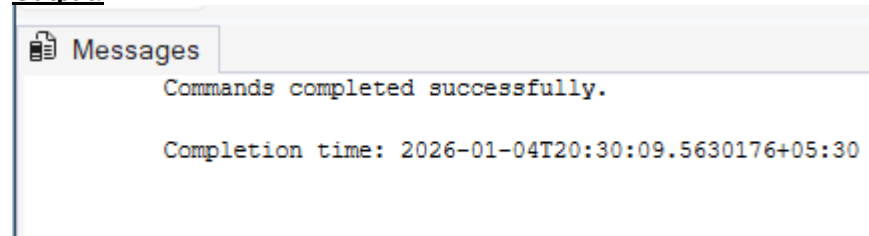
Results Messages			
	QTYORDERED	QTYDISP	BalancedQty
1	4	4	0
2	2	2	0
3	3	1	2
4	1	1	0
5	6	4	2
6	5	5	0
7	8	6	2
8	3	3	0
9	4	2	2
10	2	2	0

(II) Write Commands to do following:

1. Make Client_no as primary key in client_master.

```
--1. Make Client_no as primary key in client_master.  
ALTER TABLE CLIENT_MASTERS  
ADD CONSTRAINT PK_Clients PRIMARY KEY (CLIENTNO);
```

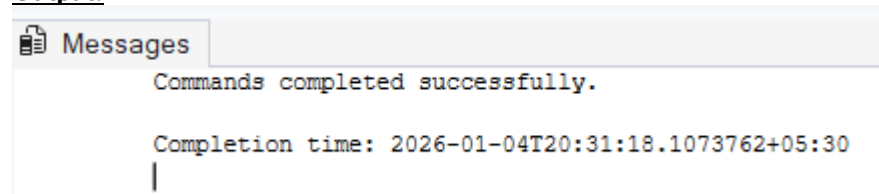
Output:



2. Add a new column phone_no in the client_master table.

```
--2. Add a new column phone_no in the client_master table.  
ALTER TABLE CLIENT_MASTER  
ADD PHONE_NO VARCHAR(11),  
CONSTRAINT chk_phone_10_digits  
CHECK (  
    LEN(Phone_No) = 10  
    AND Phone_No NOT LIKE '%[^0-9]%'  
);
```

Output:



3. Add the not null constraint in the product_master table with the column description, profit percent, sell price and cost price.

--3. Add the not null constraint in the product_master table with the column description, profit percent, sell price and cost price.

```
ALTER TABLE PRODUCT_MASTER  
ALTER COLUMN DESCRIPTION VARCHAR(15) NOT NULL;
```

```
ALTER TABLE PRODUCT_MASTER  
ALTER COLUMN PROFITPERC DECIMAL(4,2) NOT NULL;
```

```
ALTER TABLE PRODUCT_MASTER  
DROP CONSTRAINT chk_sellprice;  
ALTER TABLE PRODUCT_MASTER  
ALTER COLUMN SELLPRICE DECIMAL(8,2) NOT NULL;
```

```
ALTER TABLE PRODUCT_MASTER  
DROP CONSTRAINT chk_costprice;  
ALTER TABLE PRODUCT_MASTER  
ALTER COLUMN COSTPRICE DECIMAL(8,2) NOT NULL;
```

Output:

```
Messages  
Commands completed successfully.  
  
Completion time: 2026-01-04T20:33:13.2303595+05:30
```

4. Change size of name column to 60 in client_master table.

--4. Change size of name column to 60 in client_master table.

```
ALTER TABLE CLIENT_MASTER  
ALTER COLUMN NAME varchar(60);
```

Output:

```
Messages  
Commands completed successfully.  
  
Completion time: 2026-01-04T20:33:57.4450437+05:30
```

5. Remove pincode column from table.

--5. Remove pincode column from table.

```
ALTER TABLE CLIENT_MASTER  
DROP COLUMN PINCODE;
```

Output:

```
Messages  
Commands completed successfully.  
  
Completion time: 2026-01-04T20:34:31.1325677+05:30
```

(III) Define in 1 or 2 lines and give one example also

1. Recursive Relationship.

A recursive relationship occurs when a table has a relationship with itself.

One record in the table is related to another record in the same table.

It is used to represent hierarchical data like manager → employee.

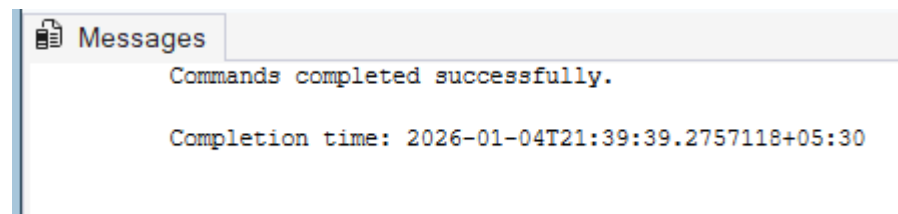
Example:

Suppose a Salesman reports to another Salesman(who is his manager)

--1. Recursive Relationship

```
ALTER TABLE SALESMAN_MASTER  
ADD ReportsTo VARCHAR(6)  
FOREIGN KEY REFERENCES SALESMAN_MASTER(SALESMANNO);
```

Output:



2. Composite Key

A composite key is a primary key formed using more than one column.

It uniquely identifies a row when a single column is not enough.

It ensures combination of fields is unique.

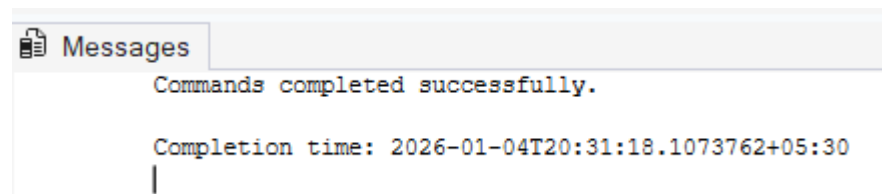
Example:

In the SALES_ORDER_DETAILS table, a composite primary key is created using ORDERNO and PRODUCTNO to uniquely identify each product within an order.

--2. Composite Key

```
ALTER TABLE SALES_ORDER_DETAILS  
ADD CONSTRAINT pk_order_product  
PRIMARY KEY (ORDERNO, PRODUCTNO);
```

Output:



3. The 'like' operator with pattern matching.

The LIKE operator is used to search specific text patterns in a column.

It uses % for multiple characters and _ for a single character.

It helps in flexible searching instead of exact match.

Example:

An SQL query to display all clients from CLIENT_MASTER whose names start with the letter 'R'.

--3. LIKE Operator

```
SELECT * FROM CLIENT_MASTER  
WHERE NAME LIKE 'R%';
```

Output:

	CLIENTNO	NAME	ADDRESS1	ADDRESS2	CITY	STATE	BALDUE	PHONE_NO
1	C00002	Ravi Kumar	B/2	Street 5	Pune	Maharashtra	12000.00	NULL
2	C00003	Riya Shah	C/7	Ring Road	Ahmedabad	Gujarat	8000.00	NULL
3	C00008	Rahul Jain	K/17	Pink Road	Jaipur	Rajasthan	14000.00	NULL

4. Drop Table command.

DROP TABLE is used to permanently delete a table from the database.

It removes the table along with all its stored data, structure, and constraints.

Once dropped, the table cannot be recovered unless a backup exists.

Example:

Deleting CLIENT_MASTERS Table

--4. DROP Table

```
DROP TABLE CLIENT_MASTERS;
```

Output:

Messages
Commands completed successfully.
Completion time: 2026-01-04T21:57:50.5478130+05:30

5. Full Outer Join

FULL OUTER JOIN Returns:

- All rows from left table
 - All rows from right table
- Even if there is **no match** between them.

If match exists → join result

If not → NULL is shown.

Example:

Show all clients and their orders (including clients without orders and orders without valid clients):

--5. FULL OUTER JOIN

```
SELECT C.CLIENTNO, C.NAME, O.ORDERNO, O.ORDERDATE
FROM CLIENT_MASTER C
FULL OUTER JOIN SALES_ORDER O
ON C.CLIENTNO = O.CLIENTNO;
```

Output:

Results		Messages		
	CLIENTNO	NAME	ORDERNO	ORDERDATE
1	C00001	Ivan Bayross	O19001	2002-06-12
2	C00002	Ravi Kumar	O19002	2002-07-15
3	C00003	Riya Shah	O19003	2002-08-01
4	C00004	Neeraj Verma	O19004	2002-08-05
5	C00006	Amit Mehta	O19006	2002-08-15
6	C00007	Priya Nair	O19007	2002-08-20
7	C00008	Rahul Jain	O19008	2002-08-25
8	C00009	Aditya Singh	O19009	2002-08-28
9	C00010	Divya Kapoor	O19010	2002-08-30
10	C00011	Arvind Kapoor	NULL	NULL

(IV) Write queries for following descriptions: (Joins)

1. Find out the products, which have been sold to 'Ivan Bayross'.

--1. Find out the products, which have been sold to 'Ivan Bayross'.

```
SELECT DISTINCT P.PRODUCTNO, P.DESRIPTION
FROM CLIENT_MASTER C
JOIN SALES_ORDER O
    ON C.CLIENTNO=O.CLIENTNO
JOIN SALES_ORDER_DETAILS D
    ON O.ORDERNO=D.ORDERNO
JOIN PRODUCT_MASTER P
    ON D.PRODUCTNO=P.PRODUCTNO
WHERE C.NAME='Ivan Bayross';
```

Output:

Results Messages		
	PRODUCTNO	DESCRIPTION
1	P00001	T-Shirts

2. Finding out the products and their quantities that will have to be delivered in the current month.

--2. Finding out the products and their quantities that will have to be delivered in the current month.

```
SELECT P.PRODUCTNO, P.DESRIPTION, COUNT(D.QTYORDERED-D.QTYDISP) AS QUANTITY
FROM PRODUCT_MASTER P
JOIN SALES_ORDER_DETAILS D
    ON P.PRODUCTNO=D.PRODUCTNO
JOIN SALES_ORDER O
    ON O.ORDERNO=D.ORDERNO
WHERE MONTH(O.DELYDATE)=MONTH(GETDATE())
AND YEAR(O.DELYDATE)=YEAR(GETDATE())
GROUP BY P.PRODUCTNO, P.DESRIPTION;
```

Output:

Results Messages			
	PRODUCTNO	DESCRIPTION	QUANTITY
1	P00001	T-Shirts	1
2	P00002	Jeans	1
3	P00003	Jackets	1
4	P00004	Shoes	1

3. Listing the ProductNo and description of constantly sold (i.e. rapidly moving) products.

--3. Listing the ProductNo and description of constantly sold (i.e. rapidly moving) products.

```
SELECT P.PRODUCTNO, P.DESCRPTION
FROM PRODUCT_MASTER P
JOIN SALES_ORDER_DETAILS D
  ON P.PRODUCTNO = D.PRODUCTNO
GROUP BY P.PRODUCTNO, P.DESCRPTION
HAVING COUNT(*)>1;
```

Output:

	PRODUCTNO	DESCRIPTION
1	P00001	T-Shirts
2	P00002	Jeans
3	P00003	Jackets

4. Finding the names of clients who have purchased 'Trousers'.

-- 4. Finding the names of clients who have purchased 'Trousers'.

```
SELECT C.NAME FROM CLIENT_MASTER C
JOIN SALES_ORDER O
  ON C.CLIENTNO=O.CLIENTNO
JOIN SALES_ORDER_DETAILS D
  ON O.ORDERNO=D.ORDERNO
JOIN PRODUCT_MASTER P
  ON D.PRODUCTNO=P.PRODUCTNO
WHERE P.DESCRPTION='Trousers';
```

Output:

	NAME
1	Ivan Bayross
2	Ravi Kumar
3	Riya Shah

5. Listing the products and orders from customers who have ordered less than 5 units of 'Pull Overs'.

--5. Listing the products and orders from customers who have ordered less than 5 units of 'Pull Overs'.

```
SELECT O.ORDERNO, C.NAME, P.DESRIPTION, SUM(D.QTYORDERED) AS QUANTITY
FROM CLIENT_MASTER C
JOIN SALES_ORDER O
    ON C.CLIENTNO=O.CLIENTNO
JOIN SALES_ORDER_DETAILS D
    ON O.ORDERNO=D.ORDERNO
JOIN PRODUCT_MASTER P
    ON D.PRODUCTNO=P.PRODUCTNO
WHERE P.DESRIPTION='Pull Overs'
GROUP BY O.ORDERNO, C.NAME, P.DESRIPTION
HAVING SUM(D.QTYORDERED)<5;
```

Output:

	ORDERNO	NAME	DESCRIPTION	QUANTITY
1	O19001	Ivan Bayross	Pull Overs	2
2	O19002	Ravi Kumar	Pull Overs	1
3	O19003	Riya Shah	Pull Overs	3

(V) Write queries for following descriptions: (Subqueries)

1. Finding the non-moving products i.e. products not being sold.

--1. Finding the non-moving products i.e. products not being sold.

```
SELECT PRODUCTNO, DESCRIPTION
FROM PRODUCT_MASTER
WHERE PRODUCTNO NOT IN (
    SELECT PRODUCTNO
    FROM SALES_ORDER_DETAILS
);
```

Output:

	PRODUCTNO	DESCRIPTION
1	P00011	1.44 drive

2. Finding the name and complete address for the customer who has placed Order number 'O19001'.

--2. Finding the name and complete address for the customer who has placed Order number 'O19001'.

```
SELECT NAME, CONCAT(ADDRESS1, ', ', ADDRESS2, ', ', CITY) AS ADDRESS
SELECT NAME, CONCAT(ADDRESS1, ', ', ADDRESS2, ', ', CITY, ', ', STATE) AS ADDRESS
FROM CLIENT_MASTER
WHERE CLIENTNO=(
    SELECT CLIENTNO
    FROM SALES_ORDER
    WHERE ORDERNO = 'O19001'
)
```

Output:

Results		Messages	
	NAME	ADDRESS	
1	Ivan Bayross	A/1,Street 1,Mumbai,Maharashtra	

3. Finding the clients who have placed orders before the month of May 2002.

--3. Finding the clients who have placed orders before the month of May'02.

```
SELECT NAME, CONCAT(ADDRESS1, ', ', ADDRESS2, ', ', CITY, ', ', STATE) AS ADDRESS
FROM CLIENT_MASTER
WHERE CLIENTNO IN (
    SELECT CLIENTNO
    FROM SALES_ORDER
    WHERE ORDERDATE < '2002-05-01'
);
```

Output:

Results		Messages	
	NAME	ADDRESS	
1	Ivan Bayross	A/1,Street 1,Mumbai,Maharashtra	
2	Ravi Kumar	B/2,Street 5,Pune,Maharashtra	
3	Riya Shah	C/7,Ring Road,Ahmedabad,Gujarat	

(VI) Write Commands to do following

1. Display system date as Saturday, February 11, 2012

--1. Display system date as Saturday, February 11, 2012

```
SELECT FORMAT(GETDATE(), 'dddd, MMMM dd, yyyy') AS SystemDate;
```

Output:

Results		Messages	
	SystemDate		
1	Sunday, January 04, 2026		

2. Display Balance Due from Client master as \$99,999.99

--2. Display Balance Due from Client master as \$99,999.99

```
SELECT NAME, FORMAT(BALDUE, '$#,##0.00') AS Balance_Due
FROM CLIENT_MASTER;
```

Output:

	NAME	Balance_Due
1	Ivan Bayross	\$15,000.00
2	Ravi Kumar	\$12,000.00
3	Riya Shah	\$8,000.00
4	Neeraj Verma	\$22,000.00
5	Amit Mehta	\$17,500.00
6	Priya Nair	\$11,000.00
7	Rahul Jain	\$14,000.00
8	Aditya Singh	\$20,000.00
9	Divya Kapoor	\$9,000.00
10	Arvind Kapoor	\$9,000.00

3. Display message as 'Salesman Aman sold goods of 50 while given target was 100.

--3. Display message as 'Salesman Aman sold goods of 50 while given target was 100.

```
SELECT 'Salesman ' + SALESMANNAME + ' sold goods of ' + CAST(YTDSALES AS
VARCHAR(10)) + ' while given target was ' + CAST(TGTTARGET AS VARCHAR(20)) AS
Message
FROM SALESMAN_MASTER
WHERE SALESMANNAME='Aman';
```

Output:

	Message
1	Salesman Aman sold goods of 50.00 while given target was 100.00

4. Display your age in years.

--4. Display your age in years.

```
SELECT DATEDIFF(YEAR, '2004-11-04', GETDATE()) AS AGE;
```

Output:

	AGE
1	22